

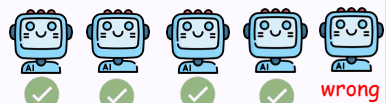
A Motivation

Diversity-Induced Under-Confidence

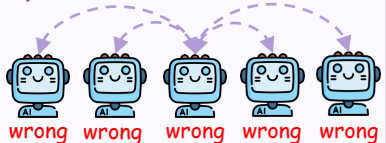


Communication-Induced Over-Confidence

1) Before communication

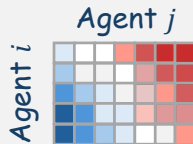


2) After communication

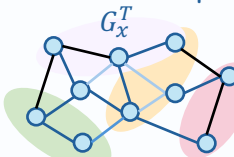


B Agent Dependency Graph Construction

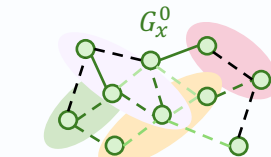
Estimate failure correlations $W(x)$



Observed Graph



Counterfactual Graph



○ Agents ■ Correlation strength
 — Communication edges
 - - - Correlation-only edges
 ● Hyperedges

Pairwise dependency edges



Group level hyperedges



C Counterfactual Shift Calibration



G_x^T



G_x^0



Calibration Agent Graph Encoder



h^T



h^0

Concatenate



h^T



h^0



$h^T - h^0$

Counterfactual Shift Calibration Learning



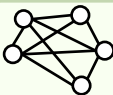
Calibrated confidence $\hat{p}(x, T)$

D CAGE-Select



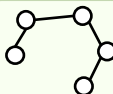
iid

$p=0.62$



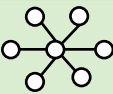
debate

$p=0.27$



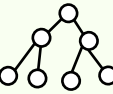
chain

$p=0.65$



hub-spoke

$p=0.88$



tree

$p=0.60$



Choose topology with highest calibrated confidence